Apurva Aggarwal

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Education

Michigan State University East Lansing, MI
Bachelor of Science in Computer Science, Minor in Business *May 2027*

- **GPA:** 3.95 — **Dean's List, 5 semesters**
- **Relevant Coursework:** Data Structures & Algorithms, Artificial Intelligence, Big Data Analysis, Cloud Development

Technical Skills

Languages: Python, SQL, C/C++, TypeScript, JavaScript, R
Data/ML: Pandas, NumPy, scikit-learn, PyTorch, PostgreSQL, SQLAlchemy
Cloud/Systems: Azure, Docker, Git, GitHub Actions, REST APIs, Flask, Next.js, CI/CD
AI/Analytics: Claude, Agents, MCP, Power BI, Tableau, Excel, KPI Design, A/B Testing

Experience

Data Analyst Intern – Tesla, Inc., Residential Energy | *Palo Alto, CA* Jan 2026 – Aug 2026

- Built a production **entity resolution pipeline** for a 35K-account installer network using fuzzy matching, indexed retrieval, and reranking; achieved **80%+ accuracy** and saved **280 hours per 10K records**
- Launched a **Power Apps CRM** for 70+ Account Managers and Sales Ops users, replacing manual Excel workflows with a visual interface and daily data refreshes; trained 30+ AMs and onboarded Sales Ops
- Built **GenAI agents** for 100+ sales and operations users across 20 workflows, connecting Jira, Teams, Confluence, and email tools for drafting messages, submitting tickets, and retrieving documentation
- Owned executive-facing **solar launch analytics**, defining KPI logic for orders, deliveries, training compliance, and kW goal attainment and automating weekly performance updates

Technology Consulting Intern – Ernst & Young (EY) | *Singapore* May 2025 – Aug 2025

- Led client-facing **user acceptance testing** for a $13M enterprise migration, validating **10K+ records** and identifying critical defects before release
- Automated migration validation with **SQL Server**, reducing manual QA effort by **50%**; triaged missing and incorrect data defects in Azure DevOps
- Helped migrate legacy files to **Azure Blob Storage** and improve access controls

Data Analytics Fellow – The Global Career Accelerator | *Remote* Aug 2024 – Dec 2024

- Analyzed **30M+ engagement records** for a Recording Academy-sponsored project, finding **5x higher engagement** for photo content; used hypothesis tests to evaluate differences at 95% confidence
- Modeled regional energy availability, demand, and **renewable capacity** for an Intel-sponsored sustainability project, ranking U.S. regions for data center readiness

Projects

Argus | *Python, Streamlit, PostgreSQL, SQLAlchemy, GitHub Actions* Jun 2026 – Present

- Built a **53-company market intelligence platform** with a **33-table PostgreSQL model** and automated pipelines for market data, SEC filings, earnings, news, and macroeconomic indicators
- Built **peer-relative company scoring** using catalysts and investment theses, with 5-, 20-, and 60-day backtests; strengthened behavioral tests and PostgreSQL-backed CI checks
- Built a local **Gemma research assistant** that translates company data into plain-language briefs, validates citation IDs, and rejects invalid responses after one repair attempt

FarmX (MHacks 2024 Winner) | *Python, scikit-learn, Streamlit* Sept 2024

- Integrated **Random Forest models** and Pandas/NumPy preprocessing into a Streamlit app for nitrogen prediction and crop recommendations using temperature, humidity, pH, and rainfall data
- Won **Best Sustainability Track at MHacks 2024** for a prototype focused on fertilizer efficiency and soil health

ISA Affiliate Platform | *Next.js, TypeScript, Supabase, PostgreSQL, Vercel* Apr 2026 – Present

- Built and deployed **msuisa.org**, a full-stack platform for affiliate operations, event registration, attendance, funding requests, and staff administration
- Implemented **server-enforced permissions** and audit records; tested organization isolation and duplicate-safe workflows, and automated releases and database migrations through CI/CD

Leadership & Involvement

SpartaHack – Co-Director for 600+ attendee student-run hackathon
International Students Association – Director of Technical Affairs; led affiliate-platform delivery
Resident Assistant – Mentored and supported 80+ students in residential community